

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I P. Karthik - 192524222 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student 

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1. Compare Bidirectional Search with standard Breadth-First Search for finding a path between a start node S and goal node G in a graph with 10 nodes. Illustrate with an example why bidirectional search can significantly reduce time complexity.
2. For a self-driving car operating on city roads, identify its percepts, possible actions, and characterize the operating environment (fully/partially observable, deterministic/stochastic, static/dynamic, discrete/continuous). Justify which agent architecture (simple reflex, model-based, goal-based, or utility-based) is most suitable and why.
3. A ride-hailing company operates a network of pickup points connected by roads with travel costs (time + fuel). Using the Uniform Cost Search (UCS) algorithm, determine the least-cost path from a starting point S to a destination G. Describe the frontier and explored set at each step for a graph of your choice with at least 6 nodes.

Compare Bidirectional search with BFS

BFS:

BFS is an uniformed search algorithm that explores all neighboring nodes level by level starting from the source node until it reaches the goal node.

Algorithm:

- 1) Start from source nodes
- 2) Add s to a queue.
- 3) Remove the front node.
- 4) Visit all unvisited neighboring nodes.

Time complexity:

• $O(b^d)$

Bidirectional search:

Bidirectional search performs two BFS searches simultaneously:

- One starts from the source (s)
- Another starts from the goal (G).

Time complexity:

• $O(b^{(d/2)})$

- Much faster than BFS for large graphs.

Ex:-



BFS search:-

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Nodes visited = 8

Bidirectional search:-

$S \rightarrow A \rightarrow D$

Backward search:-

$G \rightarrow D$

Both meet at D

Nodes visited = 4

Comparison Table:-

BFS

- Starts from source only
- Explores one direction
- More memory

Bidirectional search

- Starts from source and goal
- Explores two direction
- Less search depth

Conclusion:-

Bidirectional search is more efficient than BFS because it searches from both the start and goal simultaneously, reducing the search depth and execution time.

Self Driving car as an Intelligent Agent:

Introduction

A self driving car is an AI agent that sense its surroundings, make decisions and drives safely without human intervention.

PEAS Description

- Passenger safety
- Reach destination
- Less travel time
- Fuel efficiency.

Environment:

- Roads
- Traffic signals
- Pedestrians
- Other vehicles

Actuators

- Steering wheel
- Accelerator
- Brake
- Gear control, Horn, Indicators.

Sensors:

- Camera, GPS, LiDar, Radar, Speed sensors.

Possible actions:

- Accelerate, Break, turn, change lane, stop, overtake, park, maintain speed.

Environment characteristics:

1) Partially Observable:

The car cannot observe everything because of blind spots.

2) Stochastic:

The environment is unpredictable due to traffic and weather.

3) Dynamic:

The environment changes continuously while the car is moving.

4) Continuous:

Speed, steering angle and distance change continuously.

Advantages:

- Makes intelligent decisions
- Adapts to changing environment.
- Improves safety
- Optimizes travel time.
- Handles uncertainty.

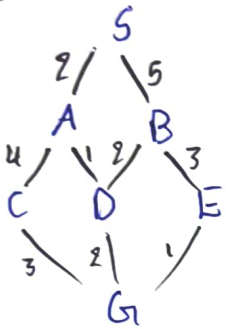
Conclusion:

A self driving car works in a partially observable, stochastic dynamic and continuous environment.

Uniform Cost search:

UCS is an informed search algorithm that always expands the node with the lowest cumulative path cost. It guarantees the least cost path when all edge costs are non-negative.

Ex:



Edge costs:

$$S \rightarrow A = 2$$

$$S \rightarrow B = 5$$

$$A \rightarrow C = 4$$

$$A \rightarrow D = 1$$

$$B \rightarrow D = 2$$

$$B \rightarrow E = 3$$

$$C \rightarrow G = 3$$

$$D \rightarrow G = 2$$

$$E \rightarrow G = 1$$

Least Cost path

$$S \rightarrow A \rightarrow D \rightarrow G$$

Total cost:

$$2 + 1 + 2 = 5$$

Frontier and Explored table

step	Frontier	Explored.
1	(S, 0)	{}
2	(A, 2) (B, 5)	S
3	(D, 3) (B, 5) (C, 6)	S, A
4	(G, 5) (B, 5) (C, 6)	S, A, D
Goal	G	S, A, D, G

Advantages:

- Finds the least cost path
- Complete algorithm
- Optimal when edge costs are non-negative.

Applications:

- Google maps
- GPS navigation
- Ride hailing services
- Robot path planning.

Conclusion

Uniform Cost search expands the node with the lowest path cost first. In this example, it finds the optimal path $s \rightarrow A \rightarrow D \rightarrow G$ with a minimum total cost of 5, making it suitable for transportation and navigation problems.